

Hermit Crab

Hardware Design Documentation

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Quadruped robotics platform · ESP32-S3 · 2S LiPo power system

This document captures the power-electronics and control-hardware design decisions behind the Hermit Crab robot. It begins with the servo motors and battery that define the system's load and energy budget, then walks through the 5 V buck converter and its component selection, the 3.3 V logic supply, the ESP32-S3 microcontroller and its USB programming interface, power source arbitration between battery and USB, and the onboard IMU and OLED display.

1. Servo Motors — MG90D Digital Micro Servo

The eight leg joints are driven by **MG90D** digital micro servos. Their peak current draw sets the sizing of the 5 V buck converter, and their PWM control scheme defines the timing requirements for the ESP32-S3.

Electrical Characteristics

- **Operating voltage:** 4.8 – 6.6 V
- **Static current:** 5 – 6 mA
- **No-load current:** 220 mA
- **Stall (blocking) current:** 800 mA + 10% $\approx$ 880 mA

PWM Control

- **Rotation direction:** CCW
- **Pulse width range:** 500 – 2500 μ s
 - **Origin position:** 500 μ s
 - **Neutral position:** 1500 μ s
 - **End position:** 2500 μ s
- **Operating frequency:** 50 – 330 Hz
 - **Standard application:** 50 Hz (20 ms PWM frame)
- The output angle is set by the high-time of the pulse — the ESP32-S3 generates one PWM channel per servo.

Impact on Power Design

Eight servos, each with an 880 mA stall current, produce a worst-case load of approximately 7.0 A. Allowing additional headroom for the ESP32 and other onboard peripherals, the 5 V rail is designed to supply up to 8 A. In practice, all eight servos are unlikely to stall simultaneously, so the typical current draw is significantly lower; however, the power rail is sized to accommodate this worst-case scenario.

2. Battery — Fytoo 2S 2000 mAh (XT30)

The robot is powered by a **Fytoo 2S 2000 mAh 15C** LiPo pack. This higher capacity compared to a smaller pack allows for significantly greater runtime, while its 15C rating still supplies the servo load with wide margin. The trade-off is added weight, which the chassis and leg servos must carry.

Specifications

- **Configuration:** 2S1P / 7.4 V nominal / 2 cell
- **Capacity:** 2000 mAh
- **Continuous discharge:** 15C
- **Discharge plug:** XT30
- **Weight:** 89g

Discharge Headroom

The C-rating sets the maximum safe current the pack can source:

- **Continuous:** $15C \times 2.0 \text{ Ah} \approx 30 \text{ A}$

This is well above the $\sim 8 \text{ A}$ worst-case system load, so the pack is never current-limited and voltage sag under a stall event is minimal.

Runtime Calculations

Stored energy at nominal voltage:

- **$2.0 \text{ Ah} \times 7.4 \text{ V} \approx 14.8 \text{ Wh}$**

Runtime is estimated as usable capacity divided by average current draw. Using 80% usable capacity (to avoid over-discharging the cells) and adding $\sim 0.3 \text{ A}$ for logic and sensors, the runtime across load scenarios is:

- **Typical gait ($\sim 400 \text{ mA/servo avg}$):** $\sim 3.5 \text{ A}$ total $\rightarrow$ roughly 27 – 28 minutes.
- **All servos no-load (220 mA each):** $\sim 2.1 \text{ A}$ total $\rightarrow$ roughly 45 minutes.
- **Worst case, all servos stalled (880 mA each):** $\sim 7.3 \text{ A}$ total $\rightarrow$ roughly 13 minutes.

3. 5 V Buck Converter — TPS54821

The main system rail is generated by a **TPS54821** synchronous step-down converter, sized to drive eight servos plus the logic electronics from a 2S LiPo battery.

Power Budget

- **Servos:** $8 \times 1 \text{ A}$ peak
- **ESP32-S3:** 500 mA peak during Wi-Fi transmission spikes
- **IMU and OLED:** negligible

Design Requirements

- **Input range:** 6 V – 8.4 V (2S LiPo)
- **Output:** 5 V @ 8 A

Datasheet: <https://www.ti.com/lit/ds/symlink/tps54821.pdf>

Reference Schematic

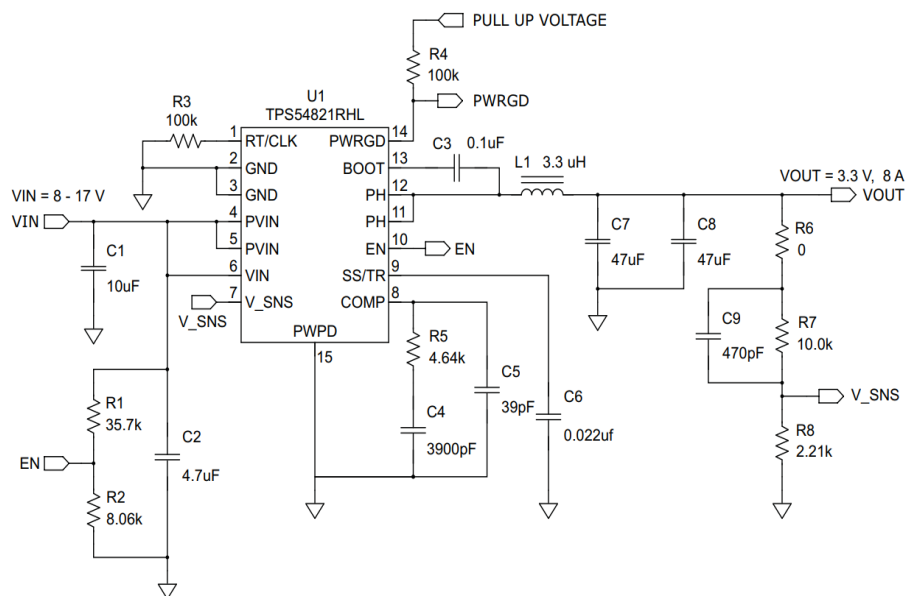


Figure 29. Typical Application Circuit

TPS54821 buck converter, modelled on the datasheet application example.

Design Summary

Parameter	Value
Output voltage (Vout)	5 V
Output current	8 A
Transient response, 4 A load step (ΔI_{out})	$\Delta V_{out} = 0.35 \text{ V}$ (7%)
Input voltage (Vin)	6.0 V (empty) 7.4 V (nominal) 8.4 V (fully charged)
Reference voltage	0.6 V
Output voltage ripple	33 mV p-p
Start input voltage (rising Vin)	6.528 V
Stop input voltage (falling Vin)	6.193 V
Operating frequency	480 kHz

Switching Frequency

The switching frequency sets a trade-off between size and efficiency:

- **Higher frequency** → smaller solution size, since lower inductance and output-capacitor values can be used.
- **Lower frequency** → higher efficiency and less heat.

A moderate **480 kHz** switching frequency was selected to balance compact size against efficient operation, matching the datasheet application example.

Reference: <https://www.ti.com/lit/wp/slyy193c/slyy193c.pdf>

Undervoltage Lockout (UVLO)

Undervoltage lockout matters for the 2S LiPo pack because over-discharging causes permanent capacity loss. The turn-on and turn-off thresholds are set by the R1/R2 divider on the EN pin.

Turn-on threshold — 6.528 V

- The converter will not begin switching until the battery voltage rises above this level.
- Prevents the converter from attempting to run when the battery is too low.

Turn-off threshold — 6.193 V

- Once running, the converter keeps switching until the battery falls below this level.
- Shuts down below the threshold to avoid undervoltage stress on the cells.

Hysteresis window

- The gap between the two thresholds stops the converter from rapidly cycling on and off when the battery hovers near the limit.
- Set by the R1/R2 divider feeding the EN pin: **R1 = 35.7 kΩ, R2 = 8.06 kΩ** (matching the application example).

Reference: <https://www.ti.com/document-viewer/lit/html/SSZTAG8>

Bootstrap Capacitor

- A **0.1 μF** ceramic capacitor (C3) is connected between the BOOT and PH pins for proper high-side gate drive.
- An X5R or better dielectric grade is recommended.

Slow-Start Capacitor

The slow-start capacitor sets the minimum time for the output to ramp to its nominal value at power-up.

For this design the soft-start time is not critical, since the $2 \times 47 \mu\text{F}$ output capacitance charges to 5 V without drawing much current. Using the TPS54821 slow-start current of **I_{ss} = 2.3 μA** and a target ramp of 6 ms:

$$C6(\text{nF}) = \frac{T_{\text{ss}}(\text{ms}) \cdot I_{\text{ss}}(\mu\text{A})}{V_{\text{ref}}(\text{V})}$$

C6 = 22 nF was selected (matching the application example).

Output Inductor

KIND is the coefficient describing inductor ripple current relative to maximum output current; typical values range from 0.1 to 0.3. A value of **0.3** was chosen here.

$$L1 = \frac{V_{\text{inmax}} - V_{\text{out}}}{I_{\text{o}} \cdot \text{Kind}} \cdot \frac{V_{\text{out}}}{V_{\text{inmax}} \cdot f_{\text{sw}}}$$

The calculated minimum is **L1 = 1.76 μH**. A larger standard **3.3 μH** value was selected to lower ripple current and match the application example.

Reference: <https://www.ti.com/lit/ta/ssztap6/ssztap6.pdf>

Output Capacitor

The output capacitor must supply load transients until the control loop responds:

- The regulator typically needs two or more clock cycles to detect a load change and adjust the duty cycle.

- The output capacitance must supply the current difference over those two cycles while limiting output voltage sag to a tolerable level.

$$C_o > \frac{2 \cdot \Delta I_{out}}{f_{sw} \cdot \Delta V_{out}}$$

The calculated minimum is **C_o = 47.6 μF**. To account for capacitance de-rating from aging, temperature, and DC bias, a conservative **2 × 47 μF** was chosen.

Input Capacitor

The TPS54821 requires a high-quality X5R or X7R ceramic decoupling capacitor of at least **4.7 μF** effective capacitance on each of the PVIN and VIN pins.

C1 = 10 μF and **C2 = 4.7 μF** were selected (matching the application example).

Output Voltage Feedback Divider

The R7/R8 divider sets the output voltage:

$$R8 = \frac{R7 \cdot V_{REF}}{V_{OUT} - V_{REF}}$$

With **R7 = 10 kΩ**, R8 calculates to 1.36 kΩ, rounded to **R8 = 1.37 kΩ** — the nearest standard 1% value.

Compensation Network

An RC network stabilizes the converter's internal error amplifier and feedback control loop.

- **Compensation resistor (R_c = R5 = 4.64 kΩ)**: in series with the compensation capacitor to the COMP pin; sets mid-band gain, loop crossover frequency, and transient response.
- **Compensation capacitor (C_c = C4 = 3900 pF)**: creates a low-frequency pole that boosts phase margin at crossover for stability.
- **High-frequency capacitor (C_p = C5 = 39 pF)**: from COMP to ground; adds a high-frequency pole to attenuate switching noise and gain spikes.
- **Feed-forward capacitor (C_{ff} = C9 = 470 pF)**: across the top feedback resistor; adds a phase-lead zero that improves phase margin and transient response and attenuates high-frequency output noise.

Because the operating conditions closely match the application example, the same compensation values were used.

4. 3.3 V Logic Supply — AP2112K-3.3TRG1

A low-dropout regulator generates the 3.3 V logic rail from the 5 V system supply.

Specifications

- **Input**: 2.5 – 6 V
- **Output**: 3.3 V @ 0.6 A
- **Enable**: tie pin 3 (EN) to VIN for normal operation (logic high).

Datasheet: <https://www.diodes.com/assets/Datasheets/AP2112.pdf>

5. Microcontroller — ESP32-S3-WROOM

The ESP32-S3-WROOM is the main controller. To keep the supply stable through power-up, an RC delay on the EN pin is recommended: **R = 10 kΩ**, **C = 1 μF**, giving roughly a 10 ms delay.

Datasheet: https://documentation.espressif.com/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf

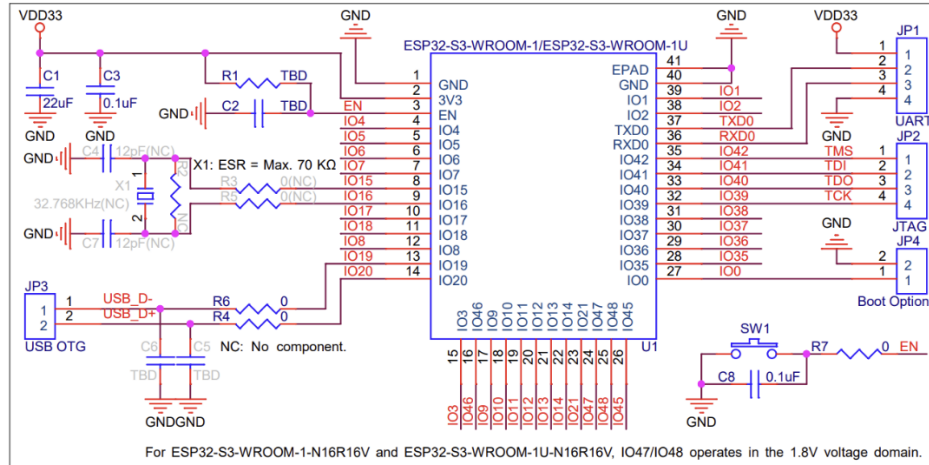


Figure 9-1. Peripheral Schematics

ESP32-S3-WROOM typical application circuit reference.

6. USB On-The-Go (OTG) Interface

The native USB-OTG interface provides serial communication and firmware flashing without an external USB-to-UART bridge.

Connection & Operation

- **Data pins:** fixed to GPIO 19 (D-) and GPIO 20 (D+).
- **USB device mode:** acts as a virtual COM port for serial communication and easy flashing.
- **Cabling:** a USB cable plugs directly into the native USB-OTG port, bypassing any external bridge chip.

Boot & Download Mode

The boot/flash header controls GPIO0, which selects the boot mode at reset:

- **GPIO0 HIGH at reset** → normal boot.
- **GPIO0 LOW at reset** → bootloader / download mode (required for flashing).

Without auto-reset circuitry, entering download mode is a manual sequence:

1. Hold BOOT.
2. Press RESET.
3. Release RESET.
4. Release BOOT.

Note: GPIO2 must also be left floating or driven low to enter the serial bootloader.

Boot mode selection: <https://docs.espressif.com/projects/esptool/en/latest/esp32/advanced-topics/boot-mode-selection.html>

USB Type-C Configuration (UFP)

- **Power direction:** the port sinks power (VBUS), drawing from the host or charger.
- **CC configuration:** a UFP applies 5.1 k Ω pull-down resistors (R_d) on both CC1 and CC2 so the system recognizes it as a device.

Reference: <https://www.ti.com/lit/wp/slyy109b/slyy109b.pdf>

Series Data Resistors

Many ESP32-S3 development boards place 22 Ω resistors between D+/D- and GPIO 19/20. These are highly recommended but not strictly required. They:

- Dampen ringing and improve signal integrity.
- Reduce EMI.
- Help with impedance matching.

7. Power Source Selection — Schottky OR-ing

Schottky OR-ing selects between USB and battery power automatically:

- **USB-C** powers the board when plugged in.
- **LiPo battery** powers the board when unplugged.
- No risk of back-feeding USB into the battery or the battery into USB.

LDO Dropout Headroom

- AP2112K dropout is typically **250 mV @ 600 mA**.
- This leaves sufficient headroom even after the Schottky diode drop.

Servo Power Routing

Note: servos must not be powered from the OR-ed 5V system rail — the worst-case 8 A load exceeds the Schottky diode rating. Instead, servos are powered directly from the 5 V buck converter output.

8. IMU — LSM6DSO32

A 6-axis IMU provides motion sensing for the platform.

Specifications

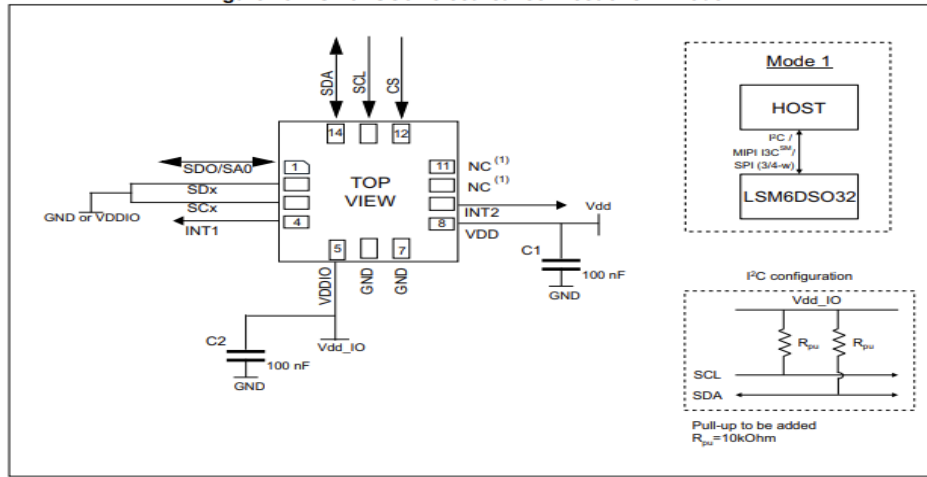
- **Supply:** 3.3 – 5 V
- **Sensors:** 3-axis accelerometer + 3-axis gyroscope.
- **Interface:** I²C — external pull-up resistors required, as the ESP32 does not provide them.

Configuration

- **SA0 (address LSB):** set to 1 (soldered).
- **CS:** set to 1 to enable I²C communication (mode 1).
- See datasheet page 24 for pin connections.

Datasheet: <https://www.st.com/resource/en/datasheet/lsm6dso32.pdf>

Figure 19. LSM6DSO32 electrical connections in Mode 1



1. Leave pin electrically unconnected and soldered to PCB.

LSM6DSO32 I²C wiring reference.

9. OLED Display — SSD1306

An SSD1306-based OLED displays Hermit's facial expressions.

Specifications

- **Supply:** 3.3 – 5 V
- **Interface:** I²C — shares the same bus as the IMU.

Datasheet: <https://cdn-shop.adafruit.com/datasheets/SSD1306.pdf>